

GASTONIA, NC, USAWMO: **723147**Lat: **35.197N**Lon: **81.156W**Elev: **243**StdP: **98.44**Time Zone: **-5.00 (NAE)**Period: **01-19**WBAN: **53870****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.0	-17.1	0.9	-1.4	-14.5	1.1	0.6	7.5	8.8	6.8	7.1	1.2	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.8	23.4	33.5	23.4	32.4	23.2	25.2	31.5	24.8	30.8	24.2	29.9	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	18.8	27.6	22.9	18.2	27.1	22.5	17.8	26.7	78.0	31.6	76.0	30.9	74.0	30.2	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.5	4.9	-10.3	36.3	2.4	2.3	-12.0	38.0	-13.4	39.3	-14.8	40.6	-16.6	42.3		
(5)			-11.3	26.4	2.2	0.6	-12.9	26.8	-14.2	27.2	-15.4	27.5	-17.0	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.6	7.5	11.7	16.5	20.6	24.8	26.4	25.9	22.9	16.8	11.1	7.8
	DBStd	8.36	5.58	5.09	5.10	4.11	3.44	2.35	1.95	2.29	3.13	4.31	4.46	4.89
	HDD10.0	453	159	101	42	4	0	0	0	0	0	3	39	104
	HDD18.3	1647	394	304	213	82	19	0	0	0	5	81	220	328
	CDD10.0	2826	24	31	94	197	330	445	507	492	386	213	72	35
	CDD18.3	979	0	1	7	26	92	195	249	234	141	32	3	1
	CDH23.3	8642	1	7	68	278	769	1774	2327	2079	1074	244	18	3
CDH26.7	3515	0	0	9	56	233	759	1072	924	406	55	1	0	
Wind	WSAvg	1.8	2.0	2.2	2.3	2.2	1.8	1.5	1.4	1.4	1.8	1.7	1.8	1.8
Precipitation	PrecAvg	1073	75	74	102	83	95	103	97	99	96	65	79	97
	PrecMax	1463	130	109	187	215	241	212	194	238	194	138	236	215
	PrecMin	715	26	30	30	23	12	30	37	18	7	0	12	26
	PrecStd	200	32	25	43	53	60	53	44	57	43	40	59	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	24.4	27.9	30.1	33.2	36.1	36.3	36.7	34.5	30.9	26.0	23.0
		MCWB	15.8	17.1	16.6	18.6	21.7	23.1	23.9	23.9	21.8	22.4	17.7	16.8
	2%	DB	19.3	21.2	25.1	28.2	30.9	33.9	34.5	34.3	32.7	27.9	22.9	20.3
		MCWB	14.6	14.2	15.8	18.0	21.1	23.2	23.9	23.6	21.9	20.1	16.2	15.9
	5%	DB	17.3	18.9	22.9	26.7	29.1	32.4	33.3	32.9	31.0	26.2	21.1	18.3
		MCWB	13.3	13.1	14.7	17.3	20.7	23.1	23.9	23.7	21.9	18.8	15.6	14.7
	10%	DB	14.7	16.5	20.6	24.7	27.6	30.9	32.1	31.4	29.0	24.0	19.0	16.2
		MCWB	10.8	11.0	13.4	16.5	20.0	22.4	23.6	23.4	21.6	18.6	14.4	12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.7	18.3	18.5	21.4	23.7	25.2	25.9	25.8	24.9	24.0	20.4	19.1
		MCDB	20.0	22.3	24.5	27.4	29.0	31.4	32.0	32.1	30.2	27.4	21.8	20.9
	2%	WB	15.8	16.5	17.3	20.1	22.6	24.5	25.3	25.2	24.0	22.4	18.5	17.4
		MCDB	18.4	19.3	22.5	25.1	28.0	30.9	31.8	31.4	28.9	25.7	21.1	19.5
	5%	WB	13.9	14.2	16.2	18.9	21.8	23.9	24.8	24.7	23.2	20.9	17.2	15.4
		MCDB	16.6	17.7	21.1	23.7	27.1	30.2	31.0	30.5	27.4	24.2	19.7	17.7
	10%	WB	11.4	11.8	15.0	17.8	21.0	23.4	24.3	24.1	22.5	19.4	15.3	13.1
		MCDB	14.1	15.2	19.1	22.6	25.9	29.2	30.1	29.4	26.6	23.0	18.5	15.4
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.4	12.1	12.6	11.6	11.1	10.7	10.5	10.5	11.7	12.2	10.7
		MCDBR	13.2	13.8	14.5	14.8	13.4	13.2	12.6	12.5	13.1	13.3	14.1	12.0
		MCWBR	9.0	8.3	7.3	6.6	5.7	4.6	4.2	4.2	4.6	6.4	8.1	7.9
	5% WB	MCDBR	10.9	11.5	12.1	11.9	11.3	11.3	10.9	10.5	10.2	10.3	10.0	10.0
		MCWBR	8.6	8.6	7.2	6.1	5.2	4.6	4.2	4.2	4.3	5.7	7.4	7.9
Clear-Sky Solar Irradiance	taub		0.313	0.321	0.360	0.400	0.455	0.482	0.530	0.514	0.425	0.381	0.332	0.314
	taud		2.498	2.469	2.396	2.277	2.161	2.125	2.015	2.045	2.279	2.381	2.481	2.527
	Ebn at Noon		893	924	911	885	837	812	769	774	832	843	861	870
	Edh at Noon		83	95	112	133	151	156	174	165	125	103	84	77
All-Sky Solar Radiation	RadAvg		2.64	3.29	4.35	5.43	6.03	6.40	5.98	5.49	4.67	3.85	2.92	2.27
	RadStd		0.15	0.38	0.31	0.42	0.49	0.46	0.40	0.26	0.42	0.47	0.24	0.21